

Detailed Summary

The project "*Neural Network & Convolutional Neural Network*" (April 2025) presents a comparative evaluation of a standard Feedforward Neural Network (NN) and a Convolutional Neural Network (CNN). The primary goal of the study was to implement, train, and test both deep learning architectures using the USPS handwritten digits dataset under a stratified 5-fold cross-validation setup to gauge model generalization.

Pre-processing Pipeline

Data was pulled from OpenML and prepared using a uniform cleaning sequence:

- **Class Filtering:** Samples labeled as "10" were completely removed because the classification target was strictly restricted to digits 0 through 9 (representing 10 distinct classes).
- **Feature Scaling:** Predictor variables were standardized utilizing StandardScaler to establish a mean of 0 and a standard deviation of 1, preventing high-scale features from skewing gradient steps.
- **Reshaping:** For the CNN pipeline alone, the flat input vectors were reshaped into 16×16 single-channel grayscale images to fit the spatial requirements of the convolutional layer.

Neural Network (NN) Model

Built using PyTorch, the feedforward baseline consisted of a 256-neuron input layer, two successive hidden layers with 128 and 64 neurons using ReLU activations, and a final 10-neuron output layer. The system was trained using CrossEntropy Loss, the Adam optimizer (set to a learning rate of 0.001), a batch size of 64, and an execution length of 20 epochs. Across the five independent cross-validation folds, the NN displayed stable convergence and reached a high average classification accuracy of $97.33\% \pm 0.0024$.

Convolutional Neural Network (CNN) Model

The CNN architecture was structurally minimized to remain lightweight for the tiny 16×16 image frames. It leveraged a single Conv2D layer containing 8 filters with a 3×3 kernel (padding=1), a ReLU activation step, and a 2×2 max-pooling operation. The resulting matrices were flattened and fed into fully connected layers scaling down from 512 → 64 → 10. Using identical parameters to the NN baseline (20 epochs, 0.001 learning rate, 64 batch size), the CNN demonstrated rapid, smooth training loss decay. It achieved a slightly superior average accuracy of $97.66\% \pm 0.0046$.

Performance Comparison & Conclusion

While both PyTorch configurations handled the USPS dataset with exceptional accuracy, the CNN systematically outpaced the dense feedforward network across the evaluation folds. The author notes that this performance discrepancy is entirely expected; the convolutional layers provide an architectural advantage by preserving spatial topologies, allowing the network to extract local geometric features like corners and edges far more effectively than flat, densely connected layers.

